AZIM PREMJI SCHOOL • CAMPUS BIODIVERSITY INITIATIVE

Campus Flora & Fauna Portal

Comprehensive System Architecture, Pedagogical Framework & Complete Feature Guide

EXECUTIVE SYSTEM INTRODUCTION

Welcome to the Azim Premji School Campus Biodiversity Portal. This document is designed for stakeholders, school evaluators, educators, and community members who have **no prior context** of this platform.

The portal is an interactive, public digital natural history registry and field guide documenting the living heritage of the school campus. It establishes equal, first-class parity between botanical **Flora** (trees, shrubs, herbs, climbers, palms, ferns, and succulents) and zoological **Fauna** (resident birds, mammals, butterflies, reptiles, amphibians, and beneficial insects). Beyond a field guide, it connects student inquiry with faculty curation through a secure **Curator & Administrator Management Portal** backed by Supabase cloud database, authentication, and image storage.

Document Purpose:	End-to-end feature specification, architecture overview, and user guide.	**Target Audience:**	School Leadership, Evaluators, Parents, Naturalists
Technology Stack:	Next.js 14 App Router, TypeScript, Supabase PostgreSQL, Storage, RLS	**Deployment Host:**	Vercel Edge Cloud + Supabase Managed DB
Release Version:	v2.0 (Dual Parity Flora & Fauna Release)	**Verification Status:**	**100% Operational (All 15 Routes Verified)**

System Navigation Gateways at a Glance

Gateway Route	Primary Audience	Core Capability & Scope	Health
/ (Home)	Public / Students	Dual-portal gateway, featured flora & fauna showcases, methodology pillars.	**Active**
/explore & /flora	Public / Botanists	Botanical directory: 10+ growth habit filters, vernacular search, comprehensive plant dossiers.	**Active**
/fauna	Public / Naturalists	Wildlife field guide: 8 taxonomic groups, activity patterns, campus wildlife dossiers.	**Active**
/about	All Visitors	4-Stage verification protocol, plant-animal interdependence, observational safety charter.	**Active**
/admin	Curator / Staff	Secure dashboard with real-time Flora/Fauna KPIs, publication pipeline feeds.	**Active**
/admin/plants	Curator / Admin	Botanical record management: creation wizard, editing, multi-photo storage, citations.	**Active**
/admin/fauna	Curator / Admin	Wildlife record management: biological data inputs, photo sync, ethics verification.	**Active**

1. Dual Biodiversity Portal (Home Page — /)

The homepage anchors the visitor experience with an elegant, natural-journal aesthetic. It communicates that the Azim Premji School campus is not merely a collection of classrooms, but a thriving ecological sanctuary where students learn through hands-on discovery.

• **Brand Identity & Global Header:** Displays school insignia, quick navigation links (Home, Flora, Fauna, About, Curator Portal), and responsive mobile drawer for smartphones and tablets.

• **Hero Section & Action Triggers:** Features the headline *'The Living Heritage of Our Campus: Flora & Fauna'* with direct one-click exploration buttons.

• **Twin Gateway Portals:** Two prominent visual cards side-by-side: *Explore Campus Flora (Botanical Field Guide)* and *Explore Campus Fauna (Wildlife Field Guide)*, highlighting live specimen counts.

• **Live Specimen Carousels:** Synchronous spotlights showing signature trees (Neem, Gulmohar) and resident creatures (Red-vented Bulbul, Palm Squirrel) with scientific binomials and verification badges.

• **Educational Inquiry Pillars:** Highlights the three core tenets: (1) Hands-On Field Observation, (2) Scientific Rigour & Taxonomy, and (3) Compassionate Campus Conservation.

• **Global Naturalist Footer:** Fast links to specific categories (e.g. Birds, Butterflies, Trees) and explicit reminders on ethical observational codes.

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2. Campus Flora Directory & Field Guide (/explore & /flora)

The Flora directory functions as a digital herbarium cataloging every verified botanical specimen. It allows students and researchers to quickly locate plants through multi-faceted search and filter tools.

• **Botanical Growth Habit Filter Pills:** Filter instantly across 10+ botanical categories: *Trees, Shrubs, Herbs, Climbers, Creepers, Flowering Plants, Grasses, Ferns, Succulents, Palms, Aquatics*.

• **Vernacular & Binomial Search:** Search query matches botanical names (*Delonix regia*), common names (Royal Poinciana), Hindi names (Gulmohar), Kannada vernaculars, or families (Fabaceae).

• **Scientific Verification Confidence Filter:** View specimens categorized by confidence: *Confirmed Specimen* (voucher verified), *Probable Identification*, or *Pending Inquiry*.

• **Interactive Plant Cards:** High-resolution photo, common name, italicized botanical binomial, family pill, campus microhabitat location, and seasonal flowering indicators.

Signature Campus Botanical Specimens Documented:



Gulmohar *(Delonix regia)*
Fabaceae • Senior Wing Courtyard



Neem *(Azadirachta indica)*
Meliaceae • Library Pathway Border



Sadabahar *(Catharanthus roseus)*
Apocynaceae • Science Block Herb Bed

Botanical Specimen Dossier Breakdown (/explore/[slug])

Each plant on campus has a comprehensive, peer-reviewed profile page providing structured scientific and ethnobotanical information. Below is the detailed breakdown of every section included on each plant's page:

Profile Section	Information Captured	Educational & Research Value
Taxonomic Profile	Kingdom, Order, Family, Genus, Species, Botanical Authority, Common Name, Hindi & Kannada vernaculars.	Teaches Linnaean classification, binomial rules, and local cultural language connections.
Morphological Traits	Bark texture & color, leaf arrangement (pinnate, alternate), flower morphology, fruit type, seed dispersal mechanism.	Trains students to identify plants based on leaf venation, bark fissures, and flower structures across seasons.
Phenology & Seasons	Flowering months, fruiting months, leaf-fall period, and seasonal growth cycles.	Enables longitudinal phenology tracking; connects seasonal climate changes to plant reproduction.
Microhabitat & Mapping	Specific campus zone (e.g. Courtyard, Sports Ground Hedges, Orchard), soil preference, sun/shade conditions.	Maps campus microclimates and serves as a guide for outdoor student botany field walks.
Ecological Importance	Pollinators attracted (bees, sunbirds, butterflies), caterpillars hosted, bird roosting value, soil enrichment.	Reveals how each tree anchors a mini-ecosystem supporting dozens of dependent organisms.
Ethnobotany & Cultural Lore	Traditional Ayurvedic/Unani medicinal usages, cultural folklore, woodwork utility, sacred associations.	Connects biological science with Indian cultural heritage and traditional knowledge systems.
Safety & Toxicity Warning	Explicit warnings regarding toxic sap (e.g. Euphorbias), poisonous seeds, dermatitis risks, and ingestion rules.	**Paramount Safety Principle:** Establishes strict rules prohibiting tasting wild seeds or rubbing sap.
Multi-Angle Photo Archive	High-resolution primary and supplementary photos (canopy, bark close-up, flower macro, fruit/seed).	Provides clear visual voucher evidence for remote verification and study.
Academic References	Authoritative sources cited (Flora of British India, Kew Royal Botanic Gardens, regional botanical bulletins).	Teaches students academic citation standards and reliance on verified literature.

CAMPUS BOTANICAL SAFETY ADVISORY

Botanical Safety Rule for Campus Explorers:
Students are guided by the cardinal rule: *'Look, examine, sketch, photograph — but never ingest.'* The portal explicitly highlights safety warnings for any species containing milky latex, cardiac glycosides, or contact irritants. Field excursions always require teacher supervision.

3. Campus Fauna Directory & Wildlife Field Guide (/fauna)

The Fauna section brings equal status to campus wildlife. It documents resident songbirds, mammals, butterflies, reptiles, amphibians, and invertebrates observed on school grounds through rigorous student field monitoring, voucher photography, and naturalist mentorship.

• **Taxonomic Class Filter Pills:** Instant 1-click sorting across 8 taxonomic groupings: *Birds (Aves), Mammals, Butterflies & Moths (Lepidoptera), Reptiles, Amphibians, Other Insects, Spiders & Arachnids, Other Invertebrates*.

• **Universal Search Engine:** Search by common name (e.g. Red-vented Bulbul), scientific binomial (*Pycnonotus cafer*), family (Agamidae), or campus hotspot (Perimeter Hedge).

• **Activity Pattern Indicators:** Badges designate each creature as **Diurnal** (active during sunlight), **Nocturnal** (active at night), or **Crepuscular** (active at twilight/dawn).

• **Observational Code of Conduct:** A permanent banner reinforces the ethics of respectful wildlife watching: safe distances, zero handling, and no nest disturbance.

• **Interactive Wildlife Cards:** Features crisp naturalist photography, common and scientific names, dietary classifications (frugivore, insectivore, omnivore), habitat notes, and verification badges.

Authentic Campus Wildlife Photography Documented in Portal:



Red-vented Bulbul
Pycnonotus cafer
Birds • Orchard



Three-striped Squirrel
Funambulus palmarum
Mammals • Courtyard



Common Mormon
Papilio polytes
Lepidoptera • Garden



Garden Lizard
Calotes versicolor
Reptiles • Perimeter

Initial Fauna Species Baseline Data:

Common / Scientific Name	Class & Family	Campus Niche & Foraging Habitat	Ecological Role & Diet
Red-vented Bulbul ***Pycnonotus cafer***	Aves Pycnonotidae	Fruit Tree Orchard & Junior Wing Lawn. Diurnal resident songbird.	Omnivorous seed disperser; controls insect pests; sips Gulmohar nectar.
Three-striped Palm Squirrel ***Funambulus palmarum***	Mammalia Sciuridae	Main Courtyard & Library Periphery canopy. Diurnal arboreal forager.	Herbivore; buries seeds that sprout into campus trees; sentinel alarm caller.
Common Mormon ***Papilio polytes***	Insecta Papilionidae	Botanical Garden & Greenhouses. Mud-puddling on moist garden soil.	Vital flower pollinator; larvae feed on Curry Leaf & Citrus bushes.
Oriental Garden Lizard ***Calotes versicolor***	Reptilia Agamidae	Perimeter Boundary Wall & sports hedges. Morning sun basking.	Insect predator (beetles, grasshoppers); shifts head color during courtship.

Zoological Specimen Dossier Breakdown (/fauna/[slug])

Each wildlife profile provides an in-depth dossier formatted specifically for field naturalists. A standout feature of this system is that it explicitly documents **Plant-Animal Ecological Interdependence**, linking wildlife directly to campus trees.

Dossier Section	Attributes & Information Recorded	Educational & Field Significance
Taxonomic Hierarchy	Order, Family, Genus, Species, Authority, English names, Hindi/regional names, IUCN Red List status.	Introduces international conservation ratings (e.g. Least Concern, Vulnerable) and global taxonomy.
Field Diagnostic Marks	Key diagnostic visual keys: crest shape, undertail colors, plumage patterns, scale keels, sexual dimorphism.	Teaches students how field naturalists separate species with precision using binoculars.
Foraging Biology & Diet	Nutritional classification (frugivorous, insectivorous, granivorous), favorite food items, hunting behaviors.	Illustrates food web dynamics and predator-prey relationships occurring right on campus.
Behaviour & Life Cycle	Nesting architecture, territorial displays, vocalization alarms, mating seasons, clutch/litter sizes, metamorphosis.	Tracks animal life stages from egg/kit to mature adult across annual school terms.
Plant-Animal Interdependence	Specific campus flora mutualisms (e.g. bulbuls pollinating Gulmohar, Common Mormon caterpillars hosting on Curry Leaf).	**Core Educational Value:** Shows students that flora and fauna form an unbroken web of mutual survival.
Campus Observation Safety Guide	Prescribed observation distances (e.g. 3–5m for birds, 2–4m for squirrels), nesting guidelines, quiet protocols.	Instills compassionate field ethics: observe without causing stress or nest abandonment.
Naturalist Discovery Notes	Direct notes from campus field walks: mud-puddling locations, sentinel warning calls, seasonal arrival dates.	Encourages students to see themselves as contributing contributors to real citizen science.

WILDLIFE OBSERVATIONAL ETHICS CHARTER

Campus Wildlife Observational Code of Conduct:

- **Respect Distance:** Maintain a quiet viewing distance of at least 3 to 5 metres for birds and 2 to 4 metres for small mammals.
- **Zero Physical Handling:** Never catch, corner, trap, or touch wild creatures on campus. Reptiles and rodents can bite when startled.
- **Nest Protection:** Never touch bird nests, fledglings on the ground, or butterfly pupae. Human scent can attract feral predators.
- **No Unnatural Feeding:** Never feed wildlife processed cafeteria snacks or bread; wild animals must maintain their natural foraging instincts.

4. Scientific Methodology & Educational Ethics (/about)

The /about page articulates the pedagogical mission of the project, explaining why Flora and Fauna are documented as an interconnected whole, and outlining the verification pipeline that ensures research-grade scientific integrity.

Why We Document Flora & Fauna Together:

Plants and animals do not live in isolation. On the Azim Premji School campus, a Gulmohar tree is not just a botanical specimen; it provides nectar to Sunbirds and Bulbuls, shelter to Palm Squirrels, and canopy shade for ground lizards. Conversely, butterflies pollinate garden herbs, while frugivorous birds disperse seeds across school lawns. Documenting them in a single portal teaches students to view the campus as a living, self-sustaining ecosystem.

The 4-Stage Scientific Verification Protocol:

Stage 1: Student Field Observation	Students observe an organism during outdoor science walks or after-school nature clubs. They record exact date/time, campus coordinates, and capture voucher photographs using classroom tablets.
Stage 2: Morphological Inquiries	Working in study groups, students compare their field observations with botanical keys and field guides, assigning an initial confidence level (Confirmed, Probable, or Uncertain).
Stage 3: Curator Review & Verification	Faculty mentors and visiting botanists review the submission in the Administrator Portal, checking diagnostic characteristics against herbaria records before approving the record to 'Published' status.
Stage 4: Phenology & Behavioral Updates	Over succeeding seasons, student cohorts update the record with new phenological observations: first bloom, fruit drop, arrival of migratory birds, and caterpillar emergence cycles.

Curriculum & Classroom Alignment:

- **Middle School Science (Grades 6–8):** Linnaean taxonomy, leaf morphology, flower anatomy, animal classification, and food web interconnectedness.
- **High School Environmental Studies (Grades 9–10):** Ecosystem services, urban biodiversity conservation, phenology as an indicator of climate change, and habitat loss.
- **Visual Arts & Botanical Illustration:** Field sketching, scientific illustration, and nature photography emphasizing diagnostic angles and scale.

5. Curator & Administrator Management Portal (/admin)

The administrative back-office gives teachers and scientific curators complete control over the portal's database. It is protected by Supabase authentication and strict Row-Level Security, ensuring that only authorized staff can add, edit, or publish records.

- **Role-Based Authentication (/login):** Secure sign-in verifying curators against an explicit admin_users table via PostgreSQL Security Definer helper functions. Unauthorized users are barred from making edits.
- **Executive KPI Analytics Dashboard (/admin):** Real-time statistics tracking: Total Documented Flora, Total Documented Fauna, Published Records, and Items Pending Review.
- **Botanical Inventory Manager (/admin/plants):** Searchable table of all plant records with quick status toggles, category filtering, and direct links to Edit or Archive.
- **Botanical Creation & Edit Wizard (/admin/plants/new & .../edit):** Comprehensive forms covering all biological parameters, multi-photo uploads to Supabase Storage, and literature references.
- **Wildlife Inventory Manager (/admin/fauna):** Management suite for campus animal records with filter by taxonomic group, verification status, and quick deletion protection.
- **Fauna Creation & Edit Wizard (/admin/fauna/new & .../edit):** Guided form capturing animal morphology, diurnal/nocturnal habits, plant mutualisms, observation distances, and multiple photographic vouchers.
- **Supabase Storage Multi-Bucket Pipeline:** Uploaded images are stored in public cloud buckets (plant-photos and fauna-photos) with automated URL linking and storage RLS protection.

Administrative Permission & Security Matrix:

Operation	Public / Student Visitor	Authenticated Curator / Admin
View Published Flora & Fauna	**Allowed (Full Access)**	**Allowed (Full Access)**
View Draft / Under Review Records	**Blocked by RLS**	**Allowed (Admin View)**
Create New Species Records	**Blocked (No Access)**	**Allowed via Form Wizard**
Upload Photos to Storage	**Blocked (Read Only)**	**Allowed (plant-photos & fauna-photos)**
Modify / Archive / Delete Records	**Blocked (Zero Mutation)**	**Allowed (Protected by is_admin())**

6. Technical Architecture & Production Verification

The platform is built on modern web standards to ensure high reliability, fast page loads, and zero downtime during campus demonstrations.

System Layer	Technology	Engineering Implementation Details
Frontend Core	Next.js 14 App Router, TypeScript, React Server Components	Optimized production bundle with static page generation (SSG) and fast dynamic routes.
Design & Styling	Vanilla CSS Design System, Custom Tokens	Handcrafted nature journal aesthetic with responsive CSS Grid; zero Tailwind bloat.
Database	Supabase PostgreSQL	Dedicated relational schemas (plants, plant_photos, fauna, fauna_photos, admin_users).
Security Model	Row Level Security (RLS) + Security Definer	public.is_admin() helper validates administrative sessions at the database engine level.
Storage Buckets	Supabase Storage (plant-photos, fauna-photos)	High-performance public image CDN with upload restrictions limited to authenticated staff.
Resilient Data Layer	Dual-Mode Repository (floraRepo.ts, faunaRepo.ts)	Syncs with live Supabase database with local fallback so pages never crash if offline.

Automated Build & Route Verification Audit:

Route / Component Tested	HTTP Status	Build / Test Result
/ (Home Portal)	**200 OK**	Static generation passed; dual banners and featured flora/fauna active.
/explore & /flora (Botanical Directory)	**200 OK**	Search, category pills, confidence filters functional.
/explore/[slug] (Plant Detail Page)	**200 OK**	Verified on *Delonix regia*; taxonomy, morphology, and warning boxes rendered.
/fauna (Wildlife Directory)	**200 OK**	Verified on 4 species; taxonomic pills, search, and activity badges functional.
/fauna/[slug] (Wildlife Detail Page)	**200 OK**	Verified on Bulbul, Squirrel, Butterfly, Lizard; field marks and safety guide active.
/about (Methodology & Safety)	**200 OK**	Dual safety charters, verification stages, and curriculum integration active.
/admin, /admin/plants, /admin/fauna	**200 OK**	Admin layout, KPI counters, and creation wizards compiled with 0 errors.

FINAL CONCLUSION

Summary & Sharing Guidance:
This portal is production-ready, deployed to Vercel, and synchronized with GitHub. Anyone with no prior knowledge can immediately navigate between Flora and Fauna, discover species living on the Azim Premji School grounds, and appreciate the scientific care embedded in its curation. For administrative onboarding or SQL database execution, refer to supabase/fauna_schema.sql.